

Cortical thickness variations and their relation to social and executive dysfunctions in autism spectrum disorder among an East Asian population: A longitudinal MRI study

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ABSTRACT

Background: The longitudinal brain structural changes and their relationships with clinical features have not been comprehensively established in individuals with autism spectrum disorder (ASD), and existing analyses have mainly focused on Western samples. This study compared change rates of cortical thickness between participants with ASD and typically developing control (TDC) and their relationships with dynamic clinical features and executive dysfunction in an East Asian sample.

Methods: We assessed 86 ASD and 82 TDC participants at two time points with structural MRI and autistic features and executive dysfunction measures. We used surface-based morphometry to identify the differences in developmental patterns between the ASD and TDC groups. We then examined the relationships between specific brain regions that vary developmentally and clinical parameters.

Results: Seven clusters were identified, showing accelerating cortical thinning in the ASD participants, including the left inferior temporal, pars orbitalis, inferior parietal, pars opercularis, right middle temporal, and bilateral rostral middle frontal regions. Greater thinning rates in the left inferior parietal and left pars opercularis regions correlated with accelerated increases in autistic features in the ASD group. Greater thinning rates in the left inferior parietal region were associated with greater executive deterioration. These structural changes also predicted social and executive dysfunction in follow-up assessments. Exploratory cross-lagged panel modeling further suggested that baseline cortical thickness in the left pars opercularis predicted social challenges at follow-up, independent of baseline autistic features. Our findings indicate that distinct developmental changes in cortical thickness are associated with social and executive dysfunction, indicating potential pseudonormalization processes that occur from adolescence to early adulthood in this population.

Conclusions: Our study elucidates neurodevelopmental correlates and trajectories of clinical deficits in ASD among Asian individuals, extending our understanding of brain-behavior relationships from adolescence to early adulthood.

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1. Introduction

Neuroimaging has played a crucial role in delineating typical and atypical neurodevelopmental trajectories. Understanding the trajectory of neuroanatomical development is fundamental to comprehending both typical cognitive and behavioral maturation (Bethlehem et al., 2022; Walhovd et al., 2014). Most neuroimaging studies have reported a wide range of regional brain morphological abnormalities in individuals with autism spectrum disorder (ASD) (Hoogman, et al., 2022). While cross-sectional studies have examined brain differences across different age groups, longitudinal studies have tracked changes in the same individuals over time with the advantage of detecting abnormalities in developmental trajectories in autistic individuals and typically developing controls (TDC) (Bieneck, et al., 2021; Mensen, et al., 2017; Pretzsch, et al., 2023; Prigge, et al., 2018, 2021; Wallace, et al., 2015; Zielinski, et al., 2014). However, few research groups have used a longitudinal design to track individual differences in the clinical course over time and their relations to brain changes (Bieneck et al., 2021; Pretzsch et al., 2023; Prigge et al., 2018; Wallace et al., 2015). Prigge et al. longitudinally assessed 91 autistic individuals and 56 TDC aged 3–39 (Prigge et al., 2018). The ASD participants with more autistic features presented regionally increased age-related cortical thinning in the left medial orbitofrontal and right inferior temporal regions. More recently, a longitudinal study followed 161 ASD individuals with subgroups based on changes in adaptive behavior: "increasing", "no change", and "decreasing". They found these clinically distinct ASD subgroups exhibited significant differences in the developmental trajectories of brain surface area and cortical thickness from baseline to follow-up (Pretzsch et al., 2023). However, some longitudinal image studies performed repeated brain imaging but only discussed the brain-behavior association cross-sectionally (Prigge et al., 2021; Wallace et al., 2015) without exploring the dynamic clinical-brain relationships in ASD. In addition, many neuroimaging studies have demonstrated that brain structures and their development might manifest differently across racial/ethnic groups (Chee et al., 2011; Choi, et al., 2020; LeWinn et al., 2017; Pfefferbaum, et al., 2016). However, existing analyses for structural-developmental patterns in ASD have predominantly focused on Western samples, limiting the generalizability of findings. Therefore, the investigation of enrolled underrepresented populations is crucial for comprehending the heterogeneity of ASD.

Executive function difficulties are frequently found in individuals with ASD (S. F. Chen, et al., 2016; Seng, et al., 2021) and are associated with autistic symptoms (Kenworthy et al., 2009; Reed et al., 2013), maladaptive functioning (Bertollo & Yerys, 2019; Gardiner & Iarocci, 2018; Wallace, et al., 2016), co-morbid psychiatric conditions (Wallace et al., 2016), and adverse adult outcomes (Hume et al., 2009). Prior research has indicated that autistic individuals tend to perform better on laboratory-based measures of executive functions than on tests that assess executive performance in daily activities (Powell & Voeller, 2004; Tsai et al., 2021). Therefore, obtaining information about executive functions from caregivers in everyday settings is crucial to understanding the real-world executive function deficits in autistic individuals (Schopler et al., 1995). These settings are typically less structured and can offer a more comprehensive understanding of the individual's executive function abilities.

Unlike laboratory tasks, the Behavior Rating Inventory of Executive Function (BRIEF) (Gioia et al., 2000) is a parent-reported assessment that evaluates an individual's executive functions in everyday demands. Although related studies have also indicated that real-world executive function challenges among autistic children change with age (Rosenthal, et al., 2013; Tschida & Yerys, 2022; Wallace et al., 2016), limited studies have reported the association between brain morphology and executive dysfunction in daily life. Wallace et al. (2015) found a possible relationship between executive dysfunction in daily life and cortical thickness in the ventral temporal cortex in 17 autistic individuals. Nonetheless, the relationship between developmental changes in executive dysfunction in daily life and cortical thickness remained unidentified. Understanding the progression of executive function and brain development in ASD can provide valuable insight into the dynamic nature of the disorder.

Cortical thickness is a sensitive indicator of cytoarchitectural and developmental changes (Sowell, et al., 2004; Wagstyl et al., 2015) and has been consistently implicated in ASD neurodevelopment (Wallace et al., 2010; Zielinski, et al., 2014). While previous cross-sectional studies have identified associations between brain structure and autistic symptoms (Faizo, 2022), there remains a lack of longitudinal research investigating the relationship between cortical thickness and autistic symptom trajectories, with most existing studies focusing on Western populations. Neurodevelopmental trajectories might differ across populations due to genetic background, environmental influences, and cultural factors; indeed, some studies have demonstrated population- or ethnicity-based differences in brain structure (Choi et al., 2020; Tang, et al., 2018). This underpins the importance of examining an underrepresented East Asian cohort. To our knowledge, there remains a limited research focus on East Asian samples, especially during the critical developmental transition from adolescence to young adulthood. Furthermore, previous longitudinal studies have shown the association between neuroanatomical correlates and executive function only cross-sectionally. The knowledge regarding the link between altered developmental trajectories of cortical thickness and executive dysfunction in daily life during this critical period for individuals with ASD remains elusive. The present study thus aimed to address this gap by examining the longitudinal associations between cortical thickness and autistic features as well as executive dysfunction in daily life during critical periods of brain development (i.e., from childhood through adolescence to adulthood) in a sample of Taiwanese individuals with ASD. The past brain imaging literature presents mixed findings (Wallace et al., 2015; Zielinski, et al., 2014). Therefore, we broadly hypothesized that distinctive developmental changes in cortical thickness would be present in ASD individuals compared to TDC. In addition, the alterations in developmental cortical thickness were anticipated to establish a meaningful association with the evolving landscape of autistic features and executive dysfunction experienced in the daily lives of autistic individuals. Finally, we examined the temporal relationships between brain structure and clinical features using a cross-lagged panel approach to investigate whether baseline brain measures could predict future changes in behavior and vice versa.

2. Methods

2.1. Participants and procedures

This study's autistic individuals were clinically diagnosed with ASD using the DSM-IV-TR (Diagnostic and Statistical Manual of Mental Disorders, 4th Edition, Text Revision) and DSM-5. The diagnoses were confirmed by the Autism Diagnostic Interview-Revised (Gau, et al., 2011) and the Autism Diagnostic Observation Schedule (Chang, et al., 2023). In addition, the psychiatric diagnoses were evaluated by the Chinese version of the Kiddie-Schedule for Affective Disorders and Schizophrenia for School-Age Children–Epidemiological Version interview at baseline and follow-up (Chen et al., 2017; Gau et al., 2005). Individuals with diagnoses of schizophrenia, mood disorders, current anxiety disorders, or substance use disorders at the time of baseline assessment were excluded.

We enrolled TDC participants recruited from the same neighborhoods as the autistic participants. These TDC did not have any lifetime or current DSM-IV-TR/DSM-5 psychiatric disorders at the time of baseline or follow-up assessments. The data for this study were collected as part of a larger prospective clinical cohort study (Chang et al., 2022; Hsiao, et al., 2022). All the participants were assessed by the Wechsler Intelligence Scale for Children–3rd Edition (age <16 years) or the Wechsler Adult Intelligence Scale–3rd Edition (age ≥16 years) at baseline, and those with a full-scale intelligence quotient < 70 were excluded.

We assessed 86 autistic individuals and 82 TDC, aged 8–25 years, with brain MRI and administered the social responsiveness scale (SRS) (Gau et al., 2013) for autistic features and the BRIEF (Tai & Gau, 2017) for impaired real-world executive functions at baseline and follow-up with a mean follow-up interval of 4.7 years (standard deviation [SD] = 1.7), at a mean age of 19.3 years (SD = 4.0). The autistic individuals and TDC in the study were recruited between 2009 and 2015. They underwent follow-up assessments between 2015 and 2020 at the outpatient clinics of National Taiwan University Hospital, a tertiary medical center located in northern Taiwan. We obtained written informed consent from all participants and their parents after a detailed explanation of the study's purpose and procedures. The study was approved by the Ethics Research Committee of the National Taiwan University Hospital, Taipei, Taiwan (approval number 201201006RIB, ClinicalTrials.gov number NCT01582256 for baseline; approval number 201403109RINC, ClinicalTrials.gov number NCT02233348 for follow-up).

2.2. Assessments

2.2.1. Autistic features

The Chinese SRS was applied to assess autistic features in the autistic participants and TDC. The SRS is a self-report or caregiver-report questionnaire frequently used to quantify the frequency of behaviors associated with autism (Constantino et al., 2007). It is not affected by age, race, IQ, or education level (Constantino et al., 2007). The SRS is a 65-item scale using a 4-point scale from 0 (never true) to 3 (almost always true), and a higher SRS score indicates more substantial characteristics associated with autism. The Chinese SRS has good reliability and validity, fair internal consistency (Cronbach's $\alpha = 0.94$ – 0.95), reliability (intra-class correlation [IC] = 0.75 – 0.85), and convergent validity with the Chinese Social Communication Questionnaire (Pearson's $r = 0.61$ – 0.87) (Gau et al., 2013). The Chinese SRS has been widely used to evaluate autistic features across general and clinical populations in Taiwan (Chang et al., 2022; Zhou, et al., 2017). All the participants were administered the Chinese SRS Parent form (age <18 years) or the Chinese SRS Adult form (age ≥18 years) at baseline and follow-up.

2.2.2. Executive dysfunction in daily life

We used the Chinese version of the BRIEF, an 86-item questionnaire, to measure individuals' real-world executive function impairments in daily social and behavioral contexts (Baron, 2000; Tsai et al., 2021). All the participants were administered the Chinese version of the BRIEF Parent form (age <18 years) or the Chinese version of the BRIEF Adult form (age ≥18 years) at baseline and follow-up. The BRIEF has two broader indices. The first index is the Behavioral Regulation Index, which includes Inhibit, Shift, Emotional Control, and Self-Monitor (only included in the adult version of the BRIEF). This index assesses the individual's capacity to regulate flexibility and modulate emotions and behaviors effectively through appropriate inhibitory control and self-monitoring. The second index is the Metacognition Index, which includes Initiate, Working Memory, Planning and Organizing, Organization of Materials, and Monitoring. This index assesses the ability to initiate tasks, utilize working memory effectively, plan and organize activities, self-manage tasks, and monitor performance cognitively. In addition, respondents are to report how often the participants have displayed a given behavior (e.g., “forgets what he/she was doing” or “talks at the wrong times” from the Working Memory and Inhibit scales, respectively) in the past 6 months. The BRIEF includes a 3-point scale with 0 (never), 1 (sometimes), and 2 (often), and a higher BRIEF score indicates greater perceived impairments (Gioia et al., 2000). The BRIEF has high internal consistency (alpha = 0.82 – 0.98) and test-retest reliability ($r = 0.72$ – 0.84) (Gioia et al., 2000). It has been used in the Taiwanese population (Chien et al., 2023; Hsieh et al., 2021; Tsai et al., 2021).

2.3. Image acquisition and preprocessing

T1-weighted images were obtained by a 3D Magnetization Prepared Rapid Acquisition Gradient Echo sequence on a 3-T MRI scanner (Siemens Magnetom Tim Trio, Germany) using 32-channel phased-arrayed head coil parameters: TR= 2000 ms; TE= 2.98 ms; TI= 900 ms; flip angle= 9°; FOV= $256 \times 256 \text{ mm}^3$; slice thickness= 1.0 mm; isotropic voxel size= 1 mm^3 . The images utilized in this study passed both automated quality control procedures and visual inspection conducted by JCC. We used the longitudinal processing pipeline in FreeSurfer version 6.0 to reduce the risk of potential over-regularization in longitudinal image processing (Reuter et al.,

2012) and Matlab version 9.4 (R2018a) to analyze the longitudinal surface-based morphology data. Cortical thickness maps were smoothed using a 10 mm full-width at half-maximum Gaussian kernel along the surface, consistent with standard FreeSurfer analysis protocols. Initially, the images from baseline and follow-up were processed separately using the cross-sectional pipeline (Fischl, et al., 2002, 2004). First, each participant's probabilistic, unbiased template was created for each time point. The image was then re-processed at each time point using the unbiased template. This longitudinal preprocessing stream used the unbiased template in several processing steps, such as skull stripping, Talairach transformations, and atlas registration. The latter steps can significantly increase reliability and statistical power when estimating volume segmentations (Reuter et al., 2012).

2.4. Statistical analysis

We compared the ASD and TDC groups with the chi-squared test for sex and independent sample t-tests for age, follow-up interval, and IQ. Our primary variable of interest was the rate of change (in mm/year) of the whole brain vertex-based primary cortical thickness. The rate of change is calculated as (Cortical Thickness at follow-up - Cortical Thickness at baseline) / (Age at follow-up - Age at baseline). Therefore, negative cortical thickness change rates were interpreted as thinning and positive as growth. A more negative value for the rate indicates a faster pace, or acceleration, of cortical thinning, whereas a more positive value indicates cortical thickening. We used FreeSurfer's Qdec program to perform whole-brain vertex-wise analyses comparing cortical thickness change rates between the ASD and TDC groups using a case-control framework. The model included sex and baseline age as covariates to control for potential confounding effects. We used Monte Carlo corrections for multiple comparisons correction (p-values < .05).

Cortical Thickness Change Rate = $\beta_0 + \beta_1 \text{ Group} + \beta_2 \text{ Age at Baseline} + \beta_3 \text{ Sex} + \epsilon$.

To examine brain-behavior associations, we extracted the cortical thickness change rates from each of the clusters identified as showing significant group differences in cortical thickness change rates. We then used Pearson correlation to examine the association between cortical thickness change rates and the change rates of autistic features (assessed by the SRS) and executive dysfunction in daily life (assessed by the BRIEF) within the ASD group under the correction of age at baseline, sex, total intracranial volume (TICV), and full-scale intelligence quotient (FIQ). For those brain regions with significant brain-behavior associations in the ASD group, we further examined the group by cortical thickness change rate interaction between the ASD and TDC groups. A significant interaction term would indicate that the relationship between cortical change and symptom change is moderated by diagnostic group.

Autistic features change rate = $\beta_0 + \beta_1 \text{ Group} + \beta_2 \text{ Cortical Thickness change rate} + \beta_3 (\text{Group X Cortical Thickness change rate}) + \beta_4 \text{ Age at baseline} + \beta_5 \text{ Sex} + \beta_6 \text{ TICV} + \beta_7 \text{ FIQ} + \epsilon$.

Executive dysfunction change rate = $\beta_0 + \beta_1 \text{ Group} + \beta_2 \text{ Cortical Thickness change rate} + \beta_3 (\text{Group X Cortical Thickness change rate}) + \beta_4 \text{ Age at baseline} + \beta_5 \text{ Sex} + \beta_6 \text{ TICV} + \beta_7 \text{ FIQ} + \epsilon$.

Subsequently, we utilized a linear model to investigate whether the cortical thickness change rates from the specific regions

Table 1

Demographics, clinical features, and global brain volumes between individuals with autism spectrum disorder and typically developing controls.

Mean (SD)	ASD (n = 86)	TDC (n = 82)	Significance or <i>p</i> -value		
Age at baseline (years)	14.84 (3.32) range: 9–25	14.29 (4.45) range: 8–25	0.362		
Age at follow-up (years)	19.65 (3.30) range: 13–29	18.90 (4.62) range: 11–30	0.234		
Follow-up Interval (years)	4.80 (1.95) range: 1.34–8.81	4.62 (1.50) range: 2.38–8.91	0.484		
Sex (Male/Female) (%)	79/7 (91.86/8.14)	57/25 (69.51/30.49)	< 0.001		
IQ at baseline					
Verbal IQ	101.65 (18.15)	111.56 (11.71)	< 0.001		
Performance IQ	100.32 (18.21)	111.50 (12.09)	< 0.001		
Full-scale IQ	101.53 (18.20)	111.27 (11.04)	< 0.001		
	Time 1 (Baseline)	Time 2 (Follow- up)	Time 1 (Baseline)	Time 2 (Follow- up)	Group, Time, Group by Time
Total scores of SRS (parent report)	90.26 (28.96)	91.46 (26.78)	28.1 (13.58)	25.68 (12.06)	a***
BRIEF (parent report)					
BRI	57.96 (10.71)	54.56 (11.99)	36.94 (6.12)	34.35 (6.33)	a***
MI	97.76 (16.19)	94.68 (17.92)	66.9 (14.82)	61.19 (15.19)	a***
Estimated Total Intra-Cranial Volumes (mm ³)	1482.72 (125.48)	1471.56 (124.59)	1440.98 (133.04)	1437.49 (138.6)	a*

a Significant group main effect in linear mixed model.

Abbreviations: ASD, autism spectrum disorder; TDC, typically developing controls; IQ: Intelligence Quotient; SRS, social responsiveness scale; BRIEF, behavior rating inventory of executive function; BRI, behavior regulation index; MI: Metacognition Index.

* p < 0.05, ** p < 0.01, ***p < 0.001

identified in our primary group difference analyses predicted the autistic features and executive dysfunction in daily life at follow-up, with covariates of age at baseline, sex, TICV, FIQ, and baseline SRS or BRIEF scores. The analyses used SAS 9.4 (SAS Institute, Cary, USA).

Yi (autistic features at follow-up) = $\beta_0 + \beta_1$ Cortical Thickness change rate + β_2 Age at baseline + β_3 Sex + β_4 TICV + β_5 FIQ + β_6 SRS-T baseline + ϵ .

Yi (executive dysfunction at follow-up) = $\beta_0 + \beta_1$ Cortical Thickness change rate + β_2 Age at baseline + β_3 Sex + β_4 TICV + β_5 FIQ + β_6 BRIEF baseline + ϵ .

Also, we employed an exploratory analysis utilizing the cross-lagged panel (CLP) model to investigate temporal brain-behavior relationships. The CLP model (Muetzel, et al., 2017), a variation of structural equation modeling, facilitates the exploration of simultaneous associations between two or more variables measured repeatedly. We identified the specific cortical brain regions exhibiting a significant correlation with autistic features/executive dysfunction as the regions of interest. Subsequently, we extracted cortical thickness data from these regions at baseline and follow-up time points from the neuroimaging data. We then employed this cortical thickness data in the CLP model to examine its association with autistic features/executive dysfunction (Supplementary Methods).

3. Results

3.1. Sample characteristics and clinical profiles

There were no significant group differences in baseline age or follow-up interval (Figure S1). However, there were more male participants and lower IQ scores at baseline in the ASD group than in the TDC group. At baseline and follow-up assessments, individuals with ASD exhibited higher total scores on the SRS (where higher scores indicate more pronounced autistic features) and BRIEF (where higher scores indicate greater executive function challenges) than the TDC (Table 1), without any significant time or group-by-time effects on autistic features or real-world executive functions assessed by these scales. The detailed change in autistic features, as assessed by the SRS total scores over age, is depicted in Figure S2A. The changes in executive dysfunction in daily life, as evaluated by the BRIEF over age, are illustrated in Figures S2B-1 and S2B-2 (supplementary material), respectively, for the Behavior Regulation Index and Metacognition Index. In addition, the ASD group displayed a larger TICV than the TDC. We found moderate to strong positive correlations between SRS total scores and both the Behavioral Regulation Index and the Metacognition Index at baseline ($r = 0.808$, $p < 0.001$; $r = 0.776$, $p < 0.001$, respectively), suggesting that greater autistic traits were associated with more pronounced everyday executive function impairments. These similar associations were observed at follow-up ($r = 0.851$, $p < 0.001$; $r = 0.836$, $p < 0.001$, respectively) (Table S1). These findings support the interrelationships between autistic social deficits and executive dysfunctions.

3.2. Developmental changes in cortical thickness

The vertex-wise analysis of the entire brain revealed the presence of seven distinct clusters that displayed a significant difference in cortical thickness change rates between the ASD and TDC groups. Specifically, greater cortical thickness reductions were observed in individuals with ASD compared to the TDC from baseline to follow-up. These cortical thickness reductions were predominantly localized to the left inferior temporal, left rostral middle frontal, left pars orbitalis, left inferior parietal, left pars opercularis, right rostral middle frontal, and right middle temporal regions (see Table 2, Fig. 1, and more cortical thickness details in Table S2). An exploratory quadratic model was depicted in the supplementary (Figure S3).

3.3. Correlations between change rates of cortical thickness and those of autistic features and executive dysfunction in daily life

We investigated brain-behavior correlations by targeting cortical regions that exhibited significant group differences in cortical thickness changes. In the autistic individuals, we found negative correlations between cortical thickness change rates in the left inferior

Table 2

Significant gray matter clusters with greater cortical thickness decrease in surface-based morphometry.

Region ^a	Size (mm ²)	x	y	z	Cluster wise p-value	Number of vertices
Left inferior temporal	2358.64	-52.1	-26.4	-17.5	< 0.001	3767
Left rostral middle frontal	989.00	-23.5	39.3	24.1	< 0.001	1546
Left pars orbitalis	829.63	-42.6	42.2	-11	< 0.001	1475
Left inferior parietal	594.23	-44.5	-67.1	24.3	< 0.001	1012
Left pars opercularis	244.98	-50.3	16.8	10.7	0.045	510
Right rostral middle frontal	601.93	18.1	52.3	18.9	< 0.001	792
Right middle temporal	327.03	57.7	-42.8	-9.6	0.021	459

^a Regions were identified based on the Harvard-Oxford Atlas

All clusters were corrected with multiple comparisons using family-wise error (FWE) correction (Monte Carlo, cluster-wise corrected p-values < .05).

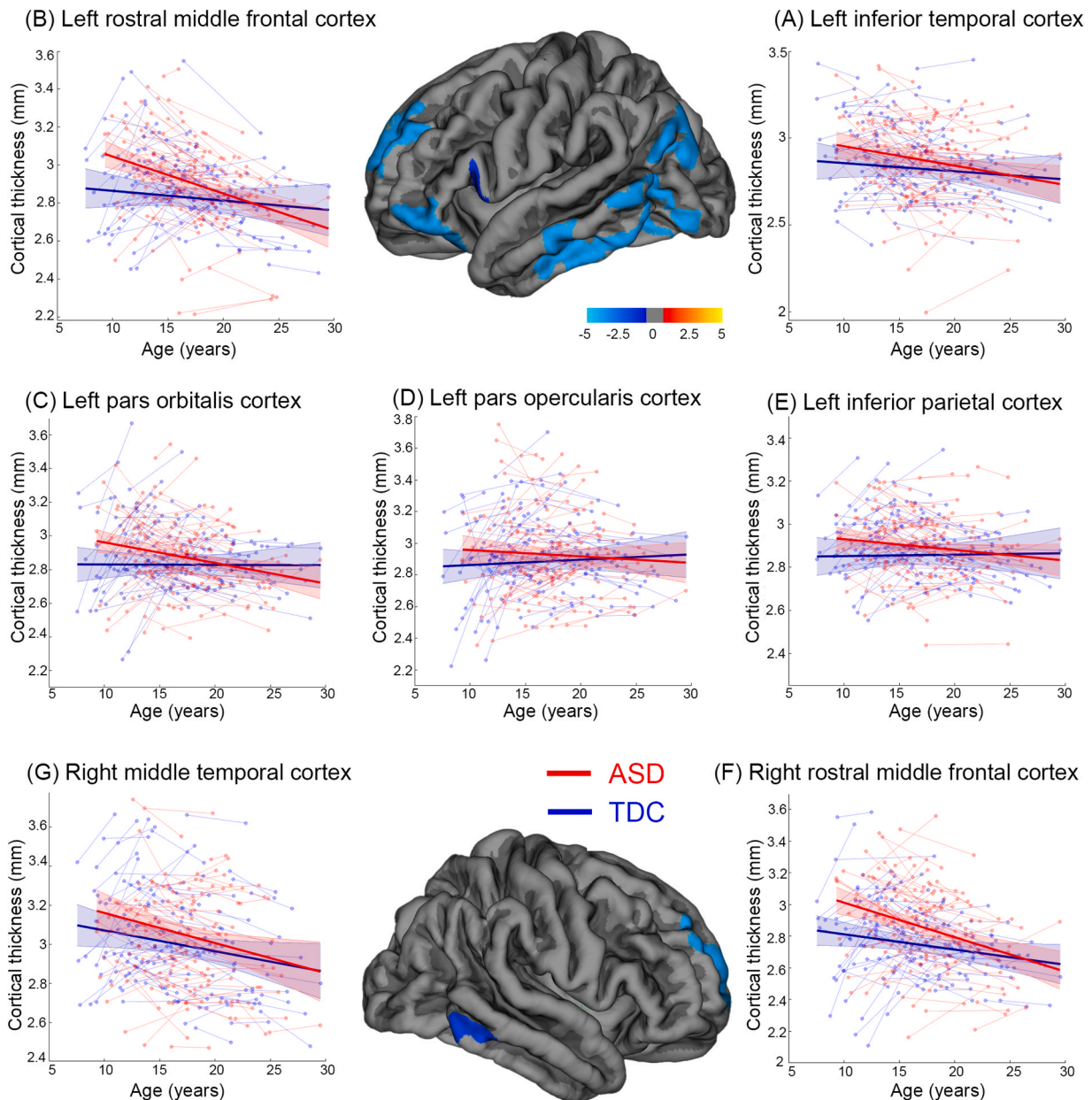


Fig. 1. Brain regions with significant differences in cortical thickness change rates between the ASD and TDC groups. Blue clusters represent areas where cortical thinning occurred more rapidly in the ASD group compared to the TDC group over the study interval. The color scale reflects z-scores derived from vertex-wise linear mixed-effects models. Significant clusters were identified using cluster-wise correction with Monte Carlo simulation ($p < 0.05$). (A) Left inferior temporal; (B) Left rostral middle frontal; (C) Left pars orbitalis; (D) Left pars opercularis; (E) Left inferior parietal; (F) Right rostral middle frontal; (G) Right middle temporal cortex. ASD, Autism Spectrum Disorder; TDC, Typically Developing Control.

parietal region and SRS total scores change rates ($r = -0.258$, $p = 0.020$, Fig. 2A) and between change rates of cortical thickness in the left pars opercularis regions and those of the SRS ($r = -0.271$, $p = 0.014$; Fig. 2B). No brain-behavior correlation was observed in the TDC group ($p > 0.05$, Fig. 2A and B). Furthermore, cortical thickness change rates in the left inferior parietal cortex showed a negative association with change rates on the BRIEF's Behavioral Regulation Index ($r = -0.246$, $p = 0.027$, Fig. 2C). The TDC group did not present an association between specific brain regions and the BRIEF ($p > 0.05$, Fig. 2C). Notably, there were significant differences in the magnitudes of the correlations between left pars opercularis thinning rates and SRS total score change rates between the ASD and TDC groups (estimate of group by cortical thickness change rate = 52.663, $p = 0.028$; Fig. 2B), and between cortical thickness change rates in the left inferior parietal cortex and Behavioral Regulation Index change rates between the two groups (estimate of group by cortical thickness change rate = 26.650, $p = 0.047$; Fig. 2C). However, no significant difference was found between the ASD and TDC

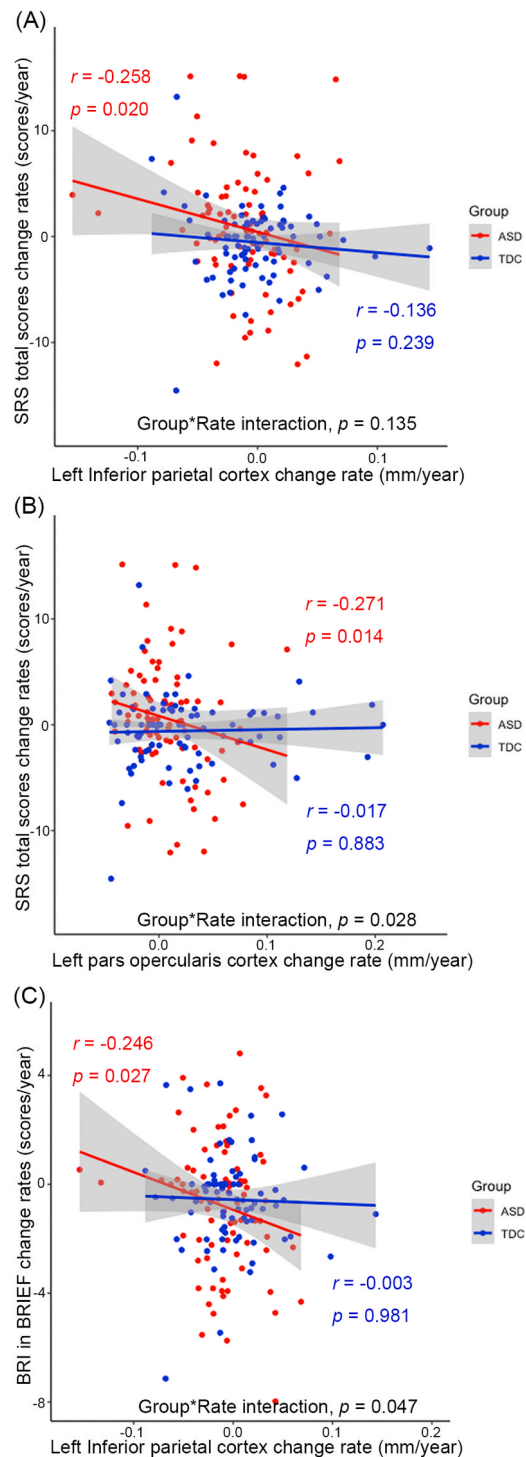


Fig. 2. Correlations of the change rate of regional cortical thickness with change rates on the SRS and BRIEF. (A) In the ASD group, a greater thinning rate of the left inferior parietal cortex correlates with an increasing SRS total score rate. No similar correlation was found in the TDC group. Therefore, there were no significant correlational differences between the ASD and TDC groups. (B) In the ASD group, a greater thinning rate of the left pars opercularis cortex was correlated with an increasing rate of the SRS total score. No similar correlation was found in the TDC group. There was a significant difference in correlation between the ASD and TDC groups. (C) In the ASD group, a greater thinning rate of the left inferior parietal cortex was correlated with an increasing rate on the BRIEF's Behavior Regulation Index. No similar correlation was found in the TDC group. There is a significant difference in correlation between the ASD and TDC groups. SRS, the Social Responsiveness Scale; BRIEF, the Behavior Rating Inventory of Executive Function; ASD, autism spectrum disorder (in red); TDC, typically developing control (in blue).

groups in the correlations between left inferior parietal thinning rates and SRS total score change rates ($p > 0.05$; Fig. 2A, detailed in Table S3).

3.4. Prediction of autistic features and executive dysfunction in daily life at follow-up

The cortical thickness change rates in the left inferior parietal region negatively predicted autistic features at follow-up (estimate = -150.147 , $p = 0.001$, Table 3A) while controlling for covariates such as baseline age, sex, TICV, FIQ, and baseline autistic features scores. The cortical thickness change rates in the left pars opercularis region's cortical thickness negatively predicted autistic features at follow-up (estimate = -145.658 , $p < 0.001$, Table 3B), controlling for the aforementioned parameters. Furthermore, the cortical thinning rate in the left pars opercularis was negatively predicted by impaired behavior regulation at follow-up, after controlling for baseline age, sex, TICV, FIQ, and baseline Behavioral Regulation Index scores (estimate = -47.370 , $p = 0.021$, Table 3C). In the sensitivity analysis, the predictive models were further adjusted for the follow-up interscan interval and the interaction term between baseline age and interscan interval to account for variability in time gaps and potential age-related differences in clinical trajectories. Neither interaction term reached statistical significance, and their inclusion did not affect the main effects (Supplementary Table S4A-C).

3.5. Exploratory cross-lagged panel modeling of temporal brain-behavior relationship

We found the baseline left pars opercularis thickness (in the region showing a cortical thickness change rate) positively associated with follow-up autistic features after adjusting for age at baseline, sex, TICV, FIQ, and baseline autistic features ($p = 0.033$; Table S5A; Figure S4A). That is, a greater pars opercularis thickness predicted greater autistic features at follow-up, independent of the expression of autistic features at baseline. Autistic features at baseline were not associated with cross-sectional cortical thickness or regional cortical thickness at follow-up. In the ASD group, similar results were found: Baseline left pars opercularis thickness was positively associated with follow-up autistic features ($p = 0.032$; Table S5B; Figure S4B). The CLP model provided a fair fit to the data (see Table S6 for fit indices). Furthermore, no temporal associations were observed between brain activity and behavior in the TDC group.

4. Discussion

As one of the few longitudinal studies to investigate cortical thickness changes in individuals with ASD and their relationships with clinical measures in an East Asian population, our results provide new insights into the distinct developmental changes in cortical thickness between ASD individuals and TDC from childhood through adolescence to early adulthood. First, accelerated cortical thinning was observed in seven regions in the ASD participants during late adolescence to early adulthood, compared with the TDC. Second, greater thinning rates on the left inferior parietal and pars opercularis regions were associated with more severe autistic features. Greater thinning rates in the left inferior parietal cortex were associated with increased deficits in inhibition, shifting, and emotional control. Third, the greater cortical thickness thinning rates of the left inferior parietal and left pars opercularis regions from childhood/early adolescence to late adolescence/early adulthood predicted more severe autistic features and executive dysfunction in daily life at follow-up, even controlling for baseline age, sex, TICV, FIQ, and corresponding clinical conditions. Fourth, an exploratory analysis using a CLP model revealed that greater pars opercularis thickness in childhood and adolescence predicted autistic features in early adulthood.

These findings of reduced cortical thickness and atypical cortical thinning with age are broadly in line with previous ASD neuroimaging studies (Ecker, et al., 2013; Hadjikhani et al., 2006; Hardan et al., 2009; Hyde et al., 2010; Jiao, et al., 2010; Mak-Fan et al., 2012; Raznahan, et al., 2010; Scheel, et al., 2011; Zielinski, et al., 2014). For example, Wallace et al. (Gregory L Wallace et al., 2010) discovered that autistic youths had a thinner temporal and parietal cortex than TDC. Additionally, they observed age-related changes in cortical thickness in certain brain regions only in the autistic individuals, specifically a thinner cortex in older autistic individuals (ages 17–24) compared to younger autistic individuals (ages 12–17). Our findings were also consistent with the very few previous longitudinal studies on Western samples (Wallace et al., 2015; Zielinski, et al., 2014). Zielinski, et al. (2014) prospectively

Table 3A

Linear regression models with left inferior parietal cortical thickness change rates as a predictor of autistic features at follow-up.

	The total score of the SRS at follow-up				
	Estimate	Standard error	Standardized Beta (β)	95 % Confidence interval	p -value
				Lower Upper	
Intercept	15.274	22.005			0.489
Age at baseline	-0.089	0.378	-0.009	-0.084 0.066	0.815
Sex	-5.385	4.244	-0.138	-0.353 0.077	0.206
FIQ	-0.108	0.098	-0.044	-0.122 0.035	0.273
Total intracranial volume	0.004	0.013	0.013	-0.071 0.097	0.754
Total score of SRS at baseline	0.848	0.041	0.838	0.758 0.918	< 0.001
Cortical thickness change rates of the left inferior parietal cortex	-150.147	45.145	-0.124	-0.198 -0.050	0.001

Abbreviations: SRS, social responsiveness scale; BRIEF, behavior rating inventory of executive function; BRI, behavior regulation index.

Table 3B

Linear regression models with left pars opercularis cortical thickness change rates as a predictor of autistic features at follow-up.

	The total score of the SRS at follow-up					<i>p</i> -value
	Estimate	Standard error	Standardized Beta (β)	95 % Confidence interval		
				Lower	Upper	
Intercept	22.642	21.966				0.304
Age at baseline	−0.335	0.381	−0.034	−0.109	0.042	0.381
Sex	−7.089	4.241	−0.182	−0.396	0.033	0.097
FIQ	−0.121	0.098	−0.049	−0.128	0.029	0.215
Total intracranial volume	0.004	0.013	0.014	−0.213	−0.063	0.744
Total score of SRS at baseline	0.836	0.041	0.826	0.746	0.906	< 0.001
Cortical thickness change rates of the left pars opercularis cortex	−145.658	40.195	−0.138	−0.070	0.097	< 0.001

Abbreviations: SRS, social responsiveness scale; BRIEF, behavior rating inventory of executive function; BRI, behavior regulation index.

Table 3C

Linear regression models with left inferior parietal cortical thickness change rates as a predictor of behavior regulation at follow-up.

	The BRI scores at follow-up					p-value
	Estimate	Standard error	Standardized Beta (β)	95 % Confidence interval		
				Lower	Upper	
Intercept	2.863	10.260				0.781
Age at baseline	0.202	0.169	0.057	-0.037	0.150	0.234
Sex	-1.159	1.919	-0.083	-0.354	0.188	0.547
FIQ	-0.032	0.044	-0.037	-0.134	0.061	0.461
Total intracranial volume	0.004	0.006	0.033	-0.073	0.138	0.539
BRI scores at baseline	0.775	0.051	0.760	0.661	0.860	< 0.001
Cortical thickness change rates of the left inferior parietal cortex	-47.370	20.369	-0.109	-0.202	-0.017	0.021

Abbreviations: SRS, social responsiveness scale; BRIEF, behavior rating inventory of executive function; BRI, behavior regulation index.

investigating 97 autistic males and 60 TDC males with an average follow-up interval of 2.6 years, found altered cortical thinning in bilateral parietal, occipital, and left frontal regions such as the pars opercularis, rostral middle frontal, and frontal pole, during the transition from adolescence to adulthood. Our findings also align with the prior longitudinal work of Pretzsch et al. (2023), which demonstrated that distinct clinical outcomes related to autistic features are associated with atypical longitudinal neurobiological profiles. This convergence suggests that some aspects of atypical cortical development in ASD may be broadly consistent across diverse populations. However, our findings did not demonstrate significant differences in the occipital region, which may be attributed to demographic variations, methodological differences, or population-specific patterns. Our study suggests a complex interplay of diverse factors that can contribute to atypical cortical development. Further research is necessary to elucidate the underlying mechanisms that govern cortical thinning and associated structural changes in autistic people.

Our findings on the associations between atypical cortical developmental patterns in the left inferior parietal and pars opercularis regions and autistic features in ASD lend support to previous studies (Hadjikhani et al., 2006; Wallace et al., 2015). The pars opercularis of the inferior frontal gyrus is among the brain regions housing a mirror neuron system, also including the inferior parietal lobule and superior temporal sulcus, and may be a potential source of the social reciprocity deficits often observed in ASD (Dapretto, et al., 2006; Molnar-Szakacs et al., 2005). Additionally, Hadjikhani et al. (2006) found that autistic individuals with thinner cortical thickness in the mirror neuron system presented with more prominent autistic features. Wallace et al. (2015) also reported a thinner cortical thickness of the ventral temporal correlated with more prominent social symptoms. Many neuroimaging experiments have shown that advanced social cognitive functions (e.g., the capacity to infer others' thoughts, beliefs, and behavioral dispositions) tend to engage the inferior parietal lobule, especially in the left hemisphere (Bzdok, et al., 2016; Bzdok, et al., 2012; Wang, et al., 2017). These findings suggest that abnormal cortical thinning in several brain regions is associated with the behavioral abnormalities commonly observed in ASD. Since the pars opercularis has been found to exhibit reduced gray matter density, cortical thinning, and abnormalities in sulcal shape in this brain region in ASD, several functional MRI studies have explored this brain region as a potential neural underpinning for the social reciprocity deficits commonly observed in ASD individuals (Hadjikhani et al., 2006; Nordahl, et al., 2007; Wang et al., 2006). Our results align with a previous study that showed age-related cortical thinning in the pars opercularis associated with higher autistic features in autistic individuals (Prigge et al., 2018). Yamasaki, et al. (2010) also found a significant gray matter volume reduction of the pars opercularis in autistic participants. The reduced volume of the pars opercularis was associated with higher social communication and reciprocity challenges in ASD.

No prior studies have used a longitudinal clinical cohort of non-Western individuals with ASD based on surface-based analysis to investigate the association between brain and cognitive presentation. From the first study, our novel finding is that autistic individuals with a greater thinning rate in the left inferior parietal region prospectively experienced worse performance in inhibition/shifting and emotional control. A cross-sectional association was identified between cortical thinning in the temporal region and parental ratings of

executive function difficulties (Wallace et al., 2015). The inferior parietal lobule is a convergence zone for various brain networks that plays a central role in enabling critical cognitive operations across different levels of the neural processing hierarchy (Kernbach, et al., 2018; Seghier, 2013). Functional MRI studies have shown that the left inferior parietal lobule is involved in various cognitive functions, including sensorimotor processing, bottom-up attention, lower-order self-perception, undirected thinking, inhibitory control, auditory perception, speech processing, and semantic processing (Wang et al., 2017). A neuroimaging study meta-analysis also found that the activity of bilateral clusters in the inferior parietal lobule was associated with scores on the Wisconsin Card Sorting Test, task switching, and Go/No-Go task (Buchsbaum et al., 2005). In short, our longitudinal finding adds evidence to support the role of the inferior parietal lobule in executive functions in ASD.

Previous cross-sectional and longitudinal studies have suggested the existence of distinct phases of cortical development in ASD (Courchesne et al., 2011; Raznahan et al., 2010; Zielinski, et al., 2014). During the initial development phase, certain cortex regions that typically undergo thinning during typical early childhood development may not thin as expected. Moreover, studies have shown an unusual increase in cortical thickness during early childhood, specifically between ages of 3 and 4 (Zielinski, et al., 2014). The early overgrowth of the cortex is followed by an overcorrection, leading to cortical thickness changes that may appear to normalize. Such pseudonormalization may occur through either earlier pruning of synapses or excessive pruning during adolescence, a stage when synapses are selectively eliminated, and dendrites and axons are developed (Huttenlocher & Dabholkar, 1997). Our data indicate that individuals with ASD experience a faster rate of cortical thinning during childhood and adolescence to early adulthood, which may indicate a pseudonormalization of cortical thickness during this stage. Additionally, the normalization of cortical thickness may not necessarily improve autistic features or executive functions, as there appeared to be an association between accelerated thinning and more severe executive dysfunctions and autistic features, suggesting that the pseudonormalization of total brain volume and cortical thickness between early childhood and young adulthood may reflect underlying cellular and molecular abnormalities, including abnormal pruning of synapses and network consolidation, that deviate from typical development. Notably, recent large-cohort studies have found continued brain overgrowth in autistic individuals rather than normalization to typical levels (Yankowitz, et al., 2020; Yankowitz et al., 2021). Our data, indicating that accelerated thinning is associated with more severe social and executive function challenges, align with the notion that such "apparent normalization" at a structural level does not equate to functional normalization. Future research should continue to investigate these complex trajectories using multi-modal approaches to understand better the cellular and functional correlates of these dynamic brain changes in ASD.

Despite the strengths of our study, some methodological limitations should be highlighted. First, the higher number of males than females prevented us from comparing the sexes in autism research, mainly due to the scarcity of female participants with ASD. Although sex was added as a covariate to reduce the possible confounding effect, caution is still necessary when interpreting the results. Future studies with larger and more evenly distributed samples of males and females are warranted to investigate any potential differences or similarities between the sexes regarding longitudinal brain structure changes. Second, our study only included verbal individuals with intelligence within the normal range ($FIQ > 70$). Hence, generalizing our results to the autistic population is questionable due to selection biases. Future structural image studies should examine cortical development in minimally verbal and lower-functioning individuals, particularly autistic adolescents and adults with intellectual disabilities. Third, some autistic individuals were receiving psychotropic agents/supplementation or psychotherapy, which may have also affected the brain. If feasible, researchers might quantify the effects of both pharmacological and non-pharmacological interventions on cortical thickness and its change across ages in future studies. Moreover, we need to consider these medications' effects in the statistical models and be cautious about interpreting the results regarding cortical thickness and ASD. Information about the relationships between cortical development and treatment interventions in individuals with ASD might help in both understanding the underlying mechanism of cortical development and developing specific measures for ASD interventions.

5. Conclusion

In summary, our study identified a unique pattern of brain development in East Asian individuals with ASD, characterized by accelerated thinning throughout adolescence into young adulthood. These brain structural alterations might play a significant role in shaping social and executive functioning in autistic people. Our findings also highlight the ongoing changes in cortical thickness and suggest a potential pseudonormalization process in ASD. Future research is warranted to explore how genetic and environmental factors influence these long-term brain changes in individuals with autism, thereby deepening our understanding of the underlying mechanisms of brain development in this population. Finally, our study revealed some brain-behavioral relationships between neurodevelopment and clinical challenges in ASD, identifying potential brain-based targets for early interventions to enhance social and executive abilities.

CRediT authorship contribution statement

Susan Shur-Fen Gau: Writing – review & editing, Project administration, Data curation, Supervision, Investigation, Conceptualization, Resources, Funding acquisition. **Yu-Chieh Chen:** Formal analysis, Conceptualization, Methodology. **Jung-Chi Chang:** Visualization, Conceptualization, Investigation, Writing – original draft, Formal analysis.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.reia.2025.202689](https://doi.org/10.1016/j.reia.2025.202689).

Data availability

Data will be made available on request.

References

- Baron, I. S. (2000). Behavior rating inventory of executive function. *Child Neuropsychology*, 6, 235–238.
- Bertollo, J. R., & Yerys, B. E. (2019). More than IQ: executive function explains adaptive behavior above and beyond nonverbal IQ in youth with autism and lower IQ. *American Journal on Intellectual and Developmental Disabilities*, 124, 191–205.
- Bethlehem, R. A. I., Seidlitz, J., White, S. R., Vogel, J. W., Anderson, K. M., Adamson, C., Adler, S., Alexopoulos, G. S., Anagnostou, E., Areces-Gonzalez, A., Astle, D. E., Auyeung, B., Ayub, M., Bae, J., Ball, G., Baron-Cohen, S., Beare, R., Bedford, S. A., Benegal, V., & Vetsa. (2022). Brain charts for the human lifespan. *Nature*, 604, 525–533.
- Bieneck, V., Bletsch, A., Mann, C., Schäfer, T., Seelemeyer, H., Herøy, N., Zimmermann, J., Pretzsch, C. M., Hattingen, E., & Ecker, C. (2021). Longitudinal changes in cortical thickness in adolescents with autism spectrum disorder and their association with restricted and repetitive behaviors. *Genes*, 12.
- Buchsbaum, B. R., Greer, S., Chang, W. L., & Berman, K. F. (2005). Meta-analysis of neuroimaging studies of the wisconsin card-sorting task and component processes. *Human Brain Mapping*, 25, 35–45.
- Bzdok, D., Hartwigsen, G., Reid, A., Laird, A. R., Fox, P. T., & Eickhoff, S. B. (2016). Left inferior parietal lobe engagement in social cognition and language. *Neuroscience & Biobehavioral Reviews*, 68, 319–334.
- Bzdok, D., Schilbach, L., Vogeley, K., Schneider, K., Laird, A. R., Langner, R., & Eickhoff, S. B. (2012). Parsing the neural correlates of moral cognition: ALE meta-analysis on morality, theory of mind, and empathy. *Brain Structure and Function*, 217, 783–796.
- Chang, J. C., Lai, M. C., Chien, Y. L., Cheng, C. Y., Wu, Y. Y., & Gau, S. S. (2023). Psychometric properties of the mandarin version of the autism diagnostic observation schedule-generic. *Journal of the Formosan Medical Association*.
- Chang, J. C., Lai, M. C., Tai, Y. M., & Gau, S. S. (2022). Mental health correlates and potential childhood predictors for the wish to be of the opposite sex in young autistic adults. *Autism*, 26, 146–159.
- Chee, M. W. L., Zheng, H., Goh, J. O. S., Park, D., & Sutton, B. P. (2011). Brain structure in young and old east asians and westerners: comparisons of structural volume and cortical thickness. *Journal of Cognitive Neuroscience*, 23, 1065–1079.
- Chen, S. F., Chien, Y. L., Wu, C. T., Shang, C. Y., Wu, Y. Y., & Gau, S. S. (2016). Deficits in executive functions among youths with autism spectrum disorders: an age-stratified analysis. *Psychological Medicine*, 46, 1625–1638.
- Chen, Y. L., Shen, L. J., & Gau, S. S. (2017). The mandarin version of the Kiddie-Schedule for affective disorders and Schizophrenia-Epidemiological version for DSM-5 - A psychometric study. *Journal of the Formosan Medical Association*, 116, 671–678.
- Chien, Y. L., Tai, Y. M., Chiu, Y. N., Tsai, W. C., & Gau, S. S. (2023). The mediators for the link between autism and real-world executive functions in adolescence and young adulthood. *Autism*, 13623613231184733.
- Choi, Y. Y., Lee, J. J., Choi, K. Y., Seo, E. H., Choo, I. H., Kim, H., Song, M. K., Choi, S. M., Cho, S. H., Kim, B. C., & Lee, K. H. (2020). The aging slopes of brain structures vary by ethnicity and sex: Evidence from a large magnetic resonance imaging dataset from a single scanner of cognitively healthy elderly people in korea. *Frontiers in Aging Neuroscience*, 12, 233.
- Constantino, J. N., Gruber, C. P., & Services, W. P. (2007). *Social responsiveness scale (SRS)*. Western Psychological Services.
- Courchesne, E., Campbell, K., & Solso, S. (2011). Brain growth across the life span in autism: Age-specific changes in anatomical pathology. *Brain Research*, 1380, 138–145.
- Dapretto, M., Davies, M. S., Pfeifer, J. H., Scott, A. A., Sigman, M., Bookheimer, S. Y., & Iacoboni, M. (2006). Understanding emotions in others: Mirror neuron dysfunction in children with autism spectrum disorders. *Nature Neuroscience*, 9, 28–30.
- Ecker, C., Gineestet, C., Feng, Y., Johnston, P., Lombardo, M. V., Lai, M. C., Suckling, J., Palaniyappan, L., Daly, E., Murphy, C. M., Williams, S. C., Bullmore, E. T., Baron-Cohen, S., Brammer, M., & Murphy, D. G. (2013). Brain surface anatomy in adults with autism: The relationship between surface area, cortical thickness, and autistic symptoms. *JAMA Psychiatry*, 70, 59–70.
- Faizo, N. L. (2022). A narrative review of MRI changes correlated to signs and symptoms of autism. *Medicine*, 101, Article e30059.
- Fischl, B., Salat, D. H., Busa, E., Albert, M., Dieterich, M., Haselgrove, C., Van Der Kouwe, A., Killiany, R., Kennedy, D., & Klaveness, S. (2002). Whole brain segmentation: Automated labeling of neuroanatomical structures in the human brain. *Neuron*, 33, 341–355.
- Fischl, B., Van Der Kouwe, A., Destrieux, C., Halgren, E., Ségonne, F., Salat, D. H., Busa, E., Seidman, L. J., Goldstein, J., & Kennedy, D. (2004). Automatically parcellating the human cerebral cortex. *Cerebral cortex*, 14, 11–22.
- Gardiner, E., & Iarocci, G. (2018). Everyday executive function predicts adaptive and internalizing behavior among children with and without autism spectrum disorder. *Autism Research*, 11, 284–295.

- Gau, S. S., Chong, M. Y., Chen, T. H., & Cheng, A. T. (2005). A 3-year panel study of mental disorders among adolescents in Taiwan. *The American Journal of Psychiatry*, 162, 1344–1350.
- Gau, S. S., Lee, C. M., Lai, M. C., Chiu, Y. N., Huang, Y. F., Kao, J. D., & Wu, Y. Y. (2011). Psychometric properties of the Chinese version of the social communication questionnaire. *Research in Autism Spectrum Disorders*, 5, 809–818.
- Gau, S. S., Liu, L. T., Wu, Y. Y., Chiu, Y. N., & Tsai, W. C. (2013). Psychometric properties of the Chinese version of the social responsiveness scale. *Research in Autism Spectrum Disorders*, 7, 349–360.
- Gioia, G. A., Isquith, P. K., Guy, S. C., & Kenworthy, L. (2000). Behavior rating inventory of executive function: BRIEF: Psychological Assessment Resources Odessa, FL.
- Hadjikhani, N., Joseph, R. M., Snyder, J., & Tager-Flusberg, H. (2006). Anatomical differences in the mirror neuron system and social cognition network in autism. *Cerebral Cortex*, 16, 1276–1282.
- Hardan, A. Y., Libove, R. A., Keshavan, M. S., Melhem, N. M., & Minshew, N. J. (2009). A preliminary longitudinal magnetic resonance imaging study of brain volume and cortical thickness in autism. *Biological Psychiatry*, 66, 320–326.
- Hoogman, M., van Rooij, D., Klein, M., Boedhoe, P., Ilioska, I., Li, T., Patel, Y., Postema, M. C., Zhang-James, Y., Anagnostou, E., Arango, C., Auzias, G., Banaschewski, T., Bau, C. H. D., Behrmann, M., Bellgrove, M. A., Brandeis, D., Brem, S., Busatto, G. F., & Franke, B. (2022). Consortium neuroscience of attention deficit/hyperactivity disorder and autism spectrum disorder: The ENIGMA adventure. *Human Brain Mapping*, 43, 37–55.
- Hsiao, M. N., Tai, Y. M., Wu, Y. Y., Tsai, W. C., Chiu, Y. N., & Gau, S. S. (2022). Psychopathologies mediate the link between autism spectrum disorder and bullying involvement: A follow-up study. *Journal of the Formosan Medical Association*, 121, 1739–1747.
- Hsieh, M. H., Chien, Y. L., & Gau, S. S. (2021). Mismatch negativity and P3a in drug-naïve adults with attention-deficit hyperactivity disorder. *Psychological Medicine*, 1–11.
- Hume, K., Loftin, R., & Lantz, J. (2009). Increasing independence in autism spectrum disorders: A review of three focused interventions. *Journal of Autism and Developmental Disorders*, 39, 1329–1338.
- Huttenlocher, P. R., & Dabholkar, A. S. (1997). Regional differences in synaptogenesis in human cerebral cortex. *Journal of Comparative Neurology*, 387, 167–178.
- Hyde, K. L., Samson, F., Evans, A. C., & Mottron, L. (2010). Neuroanatomical differences in brain areas implicated in perceptual and other core features of autism revealed by cortical thickness analysis and voxel-based morphometry. *Human Brain Mapping*, 31, 556–566.
- Jiao, Y., Chen, R., Ke, X., Chu, K., Lu, Z., & Herskovits, E. H. (2010). Predictive models of autism spectrum disorder based on brain regional cortical thickness. *Neuroimage*, 50, 589–599.
- Kenworthy, L., Black, D. O., Harrison, B., della Rosa, A., & Wallace, G. L. (2009). Are executive control functions related to autism symptoms in high-functioning children? *Child Neuropsychology*, 15, 425–440.
- Kernbach, J. M., Yeo, B. T. T., Smallwood, J., Margulies, D. S., Thiebaut de Schotten, M., Walter, H., Sabuncu, M. R., Holmes, A. J., Gramfort, A., Varoquaux, G., Thirion, B., & Bzdok, D. (2018). Subspecialization within default mode nodes characterized in 10,000 UK biobank participants. *Proceedings of the National Academy of Sciences of the United States of America*, 115, 12295–12300.
- LeWinn, K. Z., Sheridan, M. A., Keyes, K. M., Hamilton, A., & McLaughlin, K. A. (2017). Sample composition alters associations between age and brain structure. *Nature Communications*, 8, 874.
- Mak-Fan, K. M., Taylor, M. J., Roberts, W., & Lerch, J. P. (2012). Measures of cortical grey matter structure and development in children with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, 42, 419–427.
- Mensen, V. T., Wierenga, L. M., van Dijk, S., Rijks, Y., Oranje, B., Mandl, R. C., & Durston, S. (2017). Development of cortical thickness and surface area in autism spectrum disorder. *NeuroImage: Clinical*, 13, 215–222.
- Molnar-Szakacs, I., Iacoboni, M., Koski, L., & Mazziotta, J. C. (2005). Functional segregation within pars opercularis of the inferior frontal gyrus: evidence from fMRI studies of imitation and action observation. *Cerebral Cortex*, 15, 986–994.
- Muetzel, R. L., Blanken, L. M. E., van der Ende, J., El Marroun, H., Shaw, P., Sudre, G., van der Lugt, A., Jaddoe, V. W. V., Verhulst, F. C., Tiemeier, H., & White, T. (2017). Tracking brain development and dimensional psychiatric symptoms in children: A longitudinal population-based neuroimaging study. *American Journal of Psychiatry*, 175, 54–62.
- Nordahl, C. W., Dierker, D., Mostafavi, I., Schumann, C. M., Rivera, S. M., Amaral, D. G., & Van Essen, D. C. (2007). Cortical folding abnormalities in autism revealed by surface-based morphometry. *The Journal of Neuroscience*, 27, 11725–11735.
- Pfefferbaum, A., Rohlfing, T., Pohl, K. M., Lane, B., Chu, W., Kwon, D., Nolan Nichols, B., Brown, S. A., Tapert, S. F., Cummins, K., Thompson, W. K., Brumback, T., Meloy, M. J., Jernigan, T. L., Dale, A., Colrain, I. M., Baker, F. C., Prouty, D., De Bellis, M. D., Voyvodic, J. T., Clark, D. B., Luna, B., Chung, T., Nagel, B. J., & Sullivan, E. V. (2016). Adolescent development of cortical and white matter structure in the NCANDA sample: Role of sex, ethnicity, puberty, and alcohol drinking. *Cerebral Cortex*, 26, 4101–4121.
- Powell, K. B., & Voeller, K. K. (2004). Prefrontal executive function syndromes in children. *Journal of Child Neurology*, 19, 785–797.
- Pretzsch, C. M., Floris, D. L., Schäfer, T., Bletsch, A., Gurr, C., Lombardo, M. V., Chatham, C. H., Tillmann, J., Charman, T., Arenella, M., Jones, E., Ambrosino, S., Bourgeron, T., Dumas, G., Cliquet, F., Leblond, C. S., Loth, E., Oakley, B., Buitelaar, J. K., Baron-Cohen, S., Beckmann, C. F., Persico, A. M., Banaschewski, T., Durston, S., Freitag, C. M., Murphy, D. G. M., & Ecker, C. (2023). Cross-sectional and longitudinal neuroanatomical profiles of distinct clinical (adaptive) outcomes in autism. *Molecular Psychiatry*, 28, 2158–2169.
- Prigge, M. B. D., Bigler, E. D., Travers, B. G., Froehlich, A., Abildskov, T., Anderson, J. S., Alexander, A. L., Lange, N., Lainhart, J. E., & Zielinski, B. A. (2018). Social responsiveness scale (SRS) in relation to longitudinal cortical thickness changes in autism spectrum disorder. *Journal of Autism and Developmental Disorders*, 48, 3319–3329.
- Prigge, M. B. D., Lange, N., Bigler, E. D., King, J. B., Dean, D. C., 3rd, Adluru, N., Alexander, A. L., Lainhart, J. E., & Zielinski, B. A. (2021). A 16-year study of longitudinal volumetric brain development in males with autism. *Neuroimage*, 236, Article 118067.
- Raznahan, A., Toro, R., Daly, E., Robertson, D., Murphy, C., Deeley, Q., Bolton, P. F., Paus, T., & Murphy, D. G. (2010). Cortical anatomy in autism spectrum disorder: An in vivo MRI study on the effect of age. *Cerebral Cortex*, 20, 1332–1340.
- Reed, P., Watts, H., & Truzoli, R. (2013). Flexibility in young people with autism spectrum disorders on a card sort task. *Autism*, 17, 162–171.
- Reuter, M., Schmansky, N. J., Rosas, H. D., & Fischl, B. (2012). Within-subject template estimation for unbiased longitudinal image analysis. *Neuroimage*, 61, 1402–1418.
- Rosenthal, M., Wallace, G. L., Lawson, R., Wills, M. C., Dixon, E., Yerys, B. E., & Kenworthy, L. (2013). Impairments in real-world executive function increase from childhood to adolescence in autism spectrum disorders. *Neuropsychology*, 27, 13–18.
- Scheel, C., Rotarska-Jagiela, A., Schilbach, L., Lehnhardt, F. G., Krug, B., Vogeley, K., & Tepest, R. (2011). Imaging derived cortical thickness reduction in high-functioning autism: Key regions and temporal slope. *Neuroimage*, 58, 391–400.
- Schopler, E., Mesibov, G. B., & Hearsey, K. (1995). Structured teaching in the TEACCH system. *Learning and cognition in autism* (pp. 243–268). Springer.
- Seghier, M. L. (2013). The angular gyrus: Multiple functions and multiple subdivisions. *Neuroscientist*, 19, 43–61.
- Seng, G. J., Tseng, W. L., Chiu, Y. N., Tsai, W. C., Wu, Y. Y., & Gau, S. S. (2021). Executive functions in youths with autism spectrum disorder and their unaffected siblings. *Psychological Medicine*, 51, 2571–2580.
- Sowell, E. R., Thompson, P. M., Leonard, C. M., Welcome, S. E., Kan, E., & Toga, A. W. (2004). Longitudinal mapping of cortical thickness and brain growth in normal children. *The Journal of Neuroscience*, 24, 8223–8231.
- Tai, Y. M., & Gau, S. S. (2017). Depression and quality of life mediating the association between attention Deficit/Hyperactivity disorder and suicidality in military recruits. *Military Medicine*, 182, e1912–e1919.
- Tang, Y., Zhao, L., Lou, Y., Shi, Y., Fang, R., Lin, X., Liu, S., & Toga, A. (2018). Brain structure differences between Chinese and caucasian cohorts: A comprehensive morphometry study. *Human Brain Mapping*, 39, 2147–2155.
- Tsai, T. H., Chen, Y. L., & Gau, S. S. (2021). Relationships between autistic traits, insufficient sleep, and real-world executive functions in children: A mediation analysis of a national epidemiological survey. *Psychological Medicine*, 51, 579–586.
- Tschida, J. E., & Yerys, B. E. (2022). Real-world executive functioning for autistic children in school and home settings. *Autism*, 26, 1095–1107.

- Wagstyl, K., Ronan, L., Goodyer, I. M., & Fletcher, P. C. (2015). Cortical thickness gradients in structural hierarchies. *NeuroImage*, 111, 241–250.
- Walhovd, K. B., Tamnes, C. K., & Fjell, A. M. (2014). Brain structural maturation and the foundations of cognitive behavioral development. *Current Opinion in Neurology*, 27, 176–184.
- Wallace, G. L., Dankner, N., Kenworthy, L., Giedd, J. N., & Martin, A. (2010). Age-related temporal and parietal cortical thinning in autism spectrum disorders. *Brain*, 133, 3745–3754.
- Wallace, G. L., Eisenberg, I. W., Robustelli, B., Dankner, N., Kenworthy, L., Giedd, J. N., & Martin, A. (2015). Longitudinal cortical development during adolescence and young adulthood in autism spectrum disorder: Increased cortical thinning but comparable surface area changes. *Journal of the American Academy of Child and Adolescent Psychiatry*, 54, 464–469.
- Wallace, G. L., Kenworthy, L., Pugliese, C. E., Popal, H. S., White, E. I., Brodsky, E., & Martin, A. (2016). Real-World executive functions in adults with autism spectrum disorder: Profiles of impairment and associations with adaptive functioning and Co-morbid anxiety and depression. *Journal of Autism and Developmental Disorders*, 46, 1071–1083.
- Wang, A. T., Lee, S. S., Sigman, M., & Dapretto, M. (2006). Neural basis of irony comprehension in children with autism: The role of prosody and context. *Brain*, 129, 932–943.
- Wang, J., Xie, S., Guo, X., Becker, B., Fox, P. T., Eickhoff, S. B., & Jiang, T. (2017). Correspondent functional topography of the human left inferior parietal lobule at rest and under task revealed using Resting-State fMRI and coactivation based parcellation. *Human Brain Mapping*, 38, 1659–1675.
- Yamasaki, S., Yamasue, H., Abe, O., Suga, M., Yamada, H., Inoue, H., Kuwabara, H., Kawakubo, Y., Yahata, N., Aoki, S., Kano, Y., Kato, N., & Kasai, K. (2010). Reduced gray matter volume of pars opercularis is associated with impaired social communication in high-functioning autism spectrum disorders. *Biological Psychiatry*, 68, 1141–1147.
- Yankowitz, L. D., Herrington, J. D., Yerys, B. E., Pereira, J. A., Pandey, J., & Schultz, R. T. (2020). Evidence against the "normalization" prediction of the early brain overgrowth hypothesis of autism. *Molecular Autism*, 11, 51.
- Yankowitz, L. D., Yerys, B. E., Herrington, J. D., Pandey, J., & Schultz, R. T. (2021). Dissociating regional gray matter density and gray matter volume in autism spectrum condition. *Neuroimage: Clinical*, 32, Article 102888.
- Zhou, B., Zhou, H., Wu, L., Zou, X., Luo, X., Fombonne, E., Wang, Y., Yan, W., & Xu, X. (2017). Assessing the accuracy of the modified Chinese autism spectrum rating scale and social responsiveness scale for screening autism spectrum disorder in Chinese children. *Neuroscience Bulletin*, 33, 168–174.
- Zielinski, B. A., Prigge, M. B., Nielsen, J. A., Froehlich, A. L., Abildskov, T. J., Anderson, J. S., Fletcher, P. T., Zygmont, K. M., Travers, B. G., Lange, N., Alexander, A. L., Bigler, E. D., & Lainhart, J. E. (2014). Longitudinal changes in cortical thickness in autism and typical development. *Brain*, 137, 1799–1812.